Lesson 3 - Linear systems of equations

3.1 Gauss algorithm

Question 31

Level of difficulty: E

Decide whether the following system of equations is linear:

$2x_1 - 3x_2 = 5$

$3(x_1 - 2)^2 + x_2 = x_1$

Question 32

Level of difficulty: E

What are the possible solutions to a system of linear equations?

Solution question 31

Solution hint: Consider the powers of x_1 and x_2.

Short solution: No

Detailed solution: No, because x_1^2 and x_1x_2 are present. For a system of equations to be linear, the variables must be represented only to the first power and not multiplied together.

Solution question 32

Solution hint: There are three possibilities.

Solution: Exactly one solution; infinitely many solutions; no solution

Question 33

Level of difficulty: M

Based on the structure of the augmented matrix $\left(\begin{array}{cc|c} 1 & 1 & 2 \\ 2 & 2 & 3 \\ 0 & 0 & 0 \end{array}\right)$, what can be said about the solvability of the linear system of equations?

Question 34

Level of difficulty: M

Based on the structure of the following augmented matrix $\left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & 2 & 1 & 3 \end{array}\right)$, what can be said about the solvability of the linear system of equations?

Solution question 33

Solution hint: Zero lines can be omitted.

Solution: $\left(\begin{array}{cc|c} 1 & 1 & 2 \\ 2 & 2 & 3 \\ 0 & 0 & 0 \end{array}\right) \rightarrow \left(\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & 0 & -1 \end{array}\right)$. The third line, i.e., the zero line, can be omitted. After transforming the second line, you get the contradiction 0 = -1. That is, the system of equations is not solvable.

Solution question 34

Solution hint: How many unknowns and how many equations are given?

Short solution: Infinite solutions

Detailed solution: There are three unknowns and only two equations. For the system of equations to be uniquely solvable, there must be as many equations as unknowns.

Question 35

Level of difficulty: D

Give an augmented matrix with four equations and two unknowns such that the associated system of equations is uniquely solvable. Give the solution of the system of equations.

Solution question 35

Solution hint: Two equations should be able to be represented as a combination of the other two.

Short solution: $\left(\begin{array}{cc|c} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 2 & 2 & 2 \\ 2 & 4 & 6 \end{array}\right)$

Detailed Solution: $\left(\begin{array}{cc|c} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 2 & 2 & 2 \\ 2 & 4 & 6 \end{array}\right) \rightarrow \left(\begin{array}{cc|c} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right) \rightarrow \left(\begin{array}{cc|c} 1 & 1 & 1 \\ 0 & 1 & 2 \end{array}\right) \rightarrow$ $x_2 = 2$, $x_1 = -1$; several solutions possible

3.2 Solution examples with the Gauss algorithm

Question 36

Level of difficulty: E

Solve the system of linear equations $\left(\begin{array}{ccc|c} 1 & 2 & 2 & 1 \\ 2 & 1 & 2 & 0 \\ 1 & 1 & 1 & 2 \end{array}\right)$

Question 37

Level of difficulty: E

Solve the system of linear equations $\left(\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 1 & 3 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{array}\right)$

Solution question 36

Solution hint: Perform the allowed row operations so that an upper triangular matrix is formed.

Short solution: $x_1 = 3$, $x_2 = 4$, $x_3 = -5$

Detailed Solution: $\left(\begin{array}{ccc|c} 1 & 2 & 2 & 1 \\ 2 & 1 & 2 & 0 \\ 1 & 1 & 1 & 2 \end{array}\right) \rightarrow \left(\begin{array}{ccc|c} 1 & 2 & 2 & 1 \\ 0 & -3 & -2 & -2 \\ 0 & -1 & -1 & 1 \end{array}\right) \rightarrow \left(\begin{array}{ccc|c} 1 & 2 & 2 & 1 \\ 0 & -3 & -2 & -2 \\ 0 & 0 & 1 & -5 \end{array}\right)$

$\rightarrow x_3 = -5$, $-3x_2 + 10 = -2 \rightarrow x_2 = 4$, $x_1 + 8 - 10 = 1 \rightarrow x_1 = 3$

$\rightarrow$ $x_1 = 3$, $x_2 = 4$, $x_3 = -5$

Solution question 37

Solution hint: Perform the allowed row operations so that an upper triangular matrix is formed.

Solution: $\left(\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 1 & 3 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{array}\right) \rightarrow \left(\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 \end{array}\right) \rightarrow x_2 = 0$ and $x_2 = -1$. Contradiction! The system of equations is not solvable.

Question 38

Level of difficulty: M

Solve the system of linear equations $\left(\begin{array}{ccc|c} 2 & 3 & 1 & 1 \\ 1 & 1 & 2 & 2 \\ 2 & 1 & 0 & 1 \end{array}\right)$

Question 39

Level of difficulty: M

Solve the system of linear equations $\left(\begin{array}{ccc|c} 2 & 3 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 4 & 6 & 1 & 2 \end{array}\right)$

Solution question 38

Solution hint: Perform the allowed row operations so that an upper triangular matrix is formed.

Short solution: $x_1 = 5/7$, $x_2 = -3/7$, $x_3 = 6/7$

Detailed Solution: $\left(\begin{array}{ccc|c} 2 & 3 & 1 & 1 \\ 1 & 1 & 2 & 2 \\ 2 & 1 & 0 & 1 \end{array}\right) \rightarrow \left(\begin{array}{ccc|c} 2 & 3 & 1 & 1 \\ 0 & 1 & -3 & -3 \\ 0 & 2 & 1 & 0 \end{array}\right) \rightarrow \left(\begin{array}{ccc|c} 2 & 3 & 1 & 1 \\ 0 & 1 & -3 & -3 \\ 0 & 0 & 7 & 6 \end{array}\right) \rightarrow x_3 = 6/7$, $x_2 = -3/7$, $x_1 = 5/7$

Solution question 39

Solution hint: Perform the allowed row operations so that an upper triangular matrix is formed.

Short solution: $x_1 = 2$, $x_2 = -1$, $x_3 = 0$

Detailed Solution: $\left(\begin{array}{ccc|c} 2 & 3 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 4 & 6 & 1 & 2 \end{array}\right) \rightarrow \left(\begin{array}{ccc|c} 2 & 3 & 1 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & -1 & 0 \end{array}\right) \rightarrow x_3 = 0$, $x_2 = -1$, $x_1 = 2$

Question 40

Level of difficulty: D

Determine the inverse of the matrix A = $\begin{pmatrix} 1 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{pmatrix}$.

Solution question 40

Solution hint: Expand the right side with the unit matrix. Reformulate the expanded matrix so that the unit matrix is formed on the left.

Short solution: $A^{-1} = \begin{pmatrix} -\frac{1}{2} & +\frac{1}{2} & +\frac{1}{2} \\ +\frac{1}{2} & -\frac{3}{2} & +\frac{1}{2} \\ +\frac{1}{2} & +\frac{1}{2} & -\frac{1}{2} \end{pmatrix}$

Detailed solution: $\left(\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 2 & 1 & 1 & 0 & 0 & 1 \end{array}\right) \rightarrow \left(\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & -1 & 0 \\ 0 & -1 & -3 & -2 & 0 & 1 \end{array}\right)$

$\rightarrow \left(\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & -1 & 0 \\ 0 & 0 & -2 & -1 & -1 & 1 \end{array}\right) \rightarrow \left(\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & -1 & 0 \\ 0 & 0 & 1 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \end{array}\right) \rightarrow \left(\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & 0 & 0 \\ 0 & 1 & 0 & \frac{1}{2} & -\frac{3}{2} & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \end{array}\right)$

$\rightarrow \left(\begin{array}{ccc|ccc} 1 & 1 & 0 & 0 & -1 & 1 \\ 0 & 1 & 0 & \frac{1}{2} & -\frac{3}{2} & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \end{array}\right) \rightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & 1 & 0 & \frac{1}{2} & -\frac{3}{2} & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \end{array}\right)$

$\rightarrow A^{-1} = \begin{pmatrix} -\frac{1}{2} & +\frac{1}{2} & +\frac{1}{2} \\ +\frac{1}{2} & -\frac{3}{2} & +\frac{1}{2} \\ +\frac{1}{2} & +\frac{1}{2} & -\frac{1}{2} \end{pmatrix}$